// Day 12 of 30

DEEP DIVE

Java HashMap Internals — How It Actually Works

→ Swipe through all 6 slides

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What HashMap Really Is

Not just key-value. It's an array of linked lists.

```
HashMap map = new HashMap<>();

Internally:
  Node[] table;
  // Array of 16 buckets (default)
  // Each bucket = linked list/tree
```

💡 Default capacity: 16, Load factor: 0.75

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How put() Works

4 steps every time you insert

```
map.put("name", 42);

Step 1: hash("name") → hashcode
Step 2: index = hash & (n-1)
         → bucket position
Step 3: Check bucket
         Empty? → insert Node
Step 4: Not empty? → check equals()
         Same key? → overwrite
         Different? → add to chain
```

💡 hashCode() decides bucket, equals() decides match

How get() Works

Same logic, reverse direction

```
map.get("name");

Step 1: hash("name") → hashcode
Step 2: index = hash & (n-1)
Step 3: Go to bucket[index]
Step 4: Compare keys using equals()
         Found? → return value
         Not found? → return null
```

💡 O(1) average, O(n) worst case

What Happens at 75% Full

Rehashing — the expensive operation

```
When size > capacity * 0.75:
  16 * 0.75 = 12 entries triggers resize

Resize process:
  1. Create new array (32 buckets)
  2. Rehash EVERY existing entry
  3. Recompute index for each
  4. Move to new positions

// This is O(n) — expensive!
```

💡 Set initial capacity if you know the size

Treeification (Java 8+)

When a bucket gets too long

```
If one bucket has 8+ entries:
  Linked List → Red-Black Tree
  O(n) lookup → O(log n) lookup

If bucket shrinks below 6:
  Tree → back to Linked List

// Protects against hash collision attacks
```

💡 Java 8 made HashMap much more resilient

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Interview Killer Questions

You'll face these. Now you're ready.

```
Q: Why capacity is always power of 2?
A: For fast modulo: hash & (n-1)

Q: Why load factor is 0.75?
A: Balance between space and speed

Q: What if two keys have same hash?
A: Collision → same bucket → equals()

Q: Is HashMap thread-safe?
A: No. Use ConcurrentHashMap.
```

💡 These 4 questions = 90% of HashMap interviews

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```
done = true
```

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